

Evolution of Microprocessors

I. Timeline Mapping (25-points)

Instructions:

Create a concise timeline showing the evolution of microprocessors from the **Intel 4004** (1971) up to modern multi-core processors. Your timeline must include:

- At least 5 major milestones
- A short description of each (2–3 sentences)
- One key improvement introduced per generation

Guide Questions:

1. What trend do you notice in terms of processor speed and functionality?
2. How did microprocessor evolution influence modern computing devices?

Rubric Scoring

Criteria
Accuracy of Content (5 pts)
Descriptions of Milestones (5 pts)
Identification of Key Improvements (5 pts)
Quality of Insights in Guide Questions (10 pts)

II. Generations Comparison Table (20-points)

A. Fill in the table with the correct information about microprocessor generations.

Generation	Example Microprocessor	Year Introduced	Key Characteristics	Importance
1st Gen				
2nd Gen				
3rd Gen				
4th Gen				
5th Gen				

Rubric Scoring

A. Comparison Table (10 points)

Criteria
Completeness of Table (5 pts)
Accuracy of Characteristics & Importance (5 pts)

- B. **Task:** After completing the table, write a 2-3 paragraph (not sentences) comparing **3rd** and **5th** generation microprocessors. Focus on improvements in architecture and performance.

B. Comparison Paragraph (10 points)

Criteria
Depth of Comparison (4 pts)
Clarity & Organization (3 pts)
Technical Accuracy (3 pts)

III. 8086 Family Quick Analysis (30-points)

Research or recall information about the **8086-microprocessor family** and answer:

1. What addressing mode innovations were introduced by the 8086?
2. Why is the 8086 considered a “foundation” of modern x86 processors?
3. List and briefly describe **two** strengths and **two** limitations of the 8086 architectures.

Rubric Scoring

Criteria
Addressing Mode Innovations (10 pts)
Reason for Foundation of x86 (10 pts)
Strengths & Limitations (10 pts)

Overall Total Points: 75

I. Timeline Mapping

1. Intel 4004 (1971) – First Commercial Microprocessor

The Intel 4004 was the world's first commercially available microprocessor, originally designed for calculators. It was a 4-bit processor capable of performing basic arithmetic and logical operations. Although limited in speed and memory capacity, it marked the beginning of integrating a CPU into a single chip.

Key Improvement: Introduction of a single-chip CPU architecture.

References:

Intel Corporation. (n.d.). *Intel 4004 microprocessor*. Intel Museum.

<https://www.intel.com/content/www/us/en/history/museum-story-of-intel-4004.html>

2. Intel 80386 (1985) – 32-bit Processing Era

The Intel 80386 (386) expanded computing power with a full 32-bit architecture. It supported multitasking and virtual memory, which significantly improved operating system performance. This generation made personal computers more powerful and business-ready.

Key Improvement: 32-bit architecture with virtual memory support.

References:

Intel Corporation. (n.d.). *Intel 80386 processor*.

<https://www.intel.com/content/www/us/en/history/museum-story-of-intel-386.html>

3. Intel Pentium (1993) – Superscalar Architecture

The Pentium processor introduced superscalar architecture, allowing it to execute multiple instructions per clock cycle. It significantly increased speed and multimedia capabilities. This generation improved floating-point performance, benefiting graphics and gaming applications.

Key Improvement: Superscalar processing (parallel instruction execution).

References:

Intel Corporation. (n.d.). *Intel Pentium processor family*.

<https://www.intel.com/content/www/us/en/history/museum-story-of-pentium.html>

4. Intel Core i7 (2008) – Multi-Core and Hyper-Threading

The Intel Core i7 (Nehalem architecture) marked the shift toward multi-core processors becoming mainstream. It integrated memory controllers and supported

Hyper-Threading, allowing better parallel processing. This improved efficiency, multitasking, and performance for demanding applications.

Key Improvement: Multi-core processing with integrated memory controller.

References:

Intel Corporation. (2008). *Intel Core i7 processor overview*. <https://www.intel.com>

5. Apple M1 (2020) – System on Chip (SoC) & Energy Efficiency

The Apple M1 represents modern microprocessor design with ARM-based architecture and system-on-chip integration. It combines CPU, GPU, and memory into a single chip for high efficiency and performance. It demonstrates the shift toward performance-per-watt optimization in modern devices.

Key Improvement: Unified SoC design with high energy efficiency.

References:

Apple Inc. (2020). *Apple unveils M1 chip*. Apple Newsroom.
<https://www.apple.com/newsroom/2020/11/apple-unveils-m1/>

Guide Questions

1. I think the most noticeable trend is that processors became faster, smaller, and more powerful over time. In the early generations like the Intel 4004, processors were very limited and could only handle simple computations. As generations progressed, especially from the 80386 to Pentium and modern Core processors, the focus shifted not just to increasing clock speed but also to improving architecture, adding more cores, and supporting multitasking.

I also noticed that modern processors prioritize efficiency and parallel processing instead of just raw speed. Instead of increasing GHz aggressively, manufacturers improved multi-core designs and power management. This shows that performance today is more about smart design and workload distribution rather than just higher frequency.

2. I believe microprocessor evolution is the main reason why we now have powerful laptops, smartphones, and even cloud servers. If processors did not evolve from 4-bit to 64-bit multi-core systems, we would not be able to run modern operating systems, AI applications, or high-end games. Every improvement in architecture made devices more capable and more accessible to people.

In my opinion, the shift to multi-core and system-on-chip designs also made devices thinner, more energy-efficient, and portable. For example, modern chips like Apple's M1 combine CPU, GPU, and memory into one system, which improves performance while reducing power consumption. Because of these advancements, computing is no longer limited to desktops and it is now embedded in phones, smart devices, and even IoT systems

II. Generation Comparison Table

Generation	Example Microprocessor	Year Introduced	Key Characteristics	Importance
1 st Gen	Intel 4004	1971	4-bit processing, basic instruction set, limited memory	First single-chip microprocessor; foundation of modern CPUs
2 nd Gen	Intel 8086	1978	16-bit architecture, improved memory addressing, x86 instruction set	Established x86 standard used in modern computers
3 rd Gen	Intel 80386	1985	32-bit processing, virtual memory, multitasking support	Enabled advanced operating systems like GUI-based system
4 th Gen	Intel Pentium	1993	Superscalar design, improved	Boosted multimedia, gaming, and

			floating-point unit	business applications
5 th Gen	Intel Core i7	2008	Multi-core, Hyper-Threading, integrated memory controller	Supports high-performance computing and modern multitasking

B. Comparison Paragraph

- As a third-year student studying Computer Science in LPU Manila, I can observe that microprocessor evolution reflects Moore's Law in action and its transistors doubled while size decreased and performance increased. The shift from single core to multi-core processors shows that innovation is no longer just about speed, but about efficiency, parallelism, and intelligent workload distribution. These advancements directly influence fields like cybersecurity, software engineering, AI development, and cloud computing, which rely heavily on processor capability.

III. 8086 Family Quick Analysis

1.

- The 8086 introduced several advanced addressing modes compared to earlier processors like the Intel 4004 and 8080.

I think one of the biggest innovations was **segmented memory addressing**. The 8086 used segment registers (CS, DS, SS, ES) combined with 16-bit offsets to access up to 1MB of memory, even though it was only a 16-bit processor. This was innovative because it extended memory access beyond the 64KB limit of earlier CPUs.

Another major innovation was the variety of **flexible addressing modes**, including:

- Immediate addressing
- Register addressing
- Direct memory addressing
- Register indirect addressing
- Based and indexed addressing
- Based-index with displacement

I believe these modes made programming more efficient because they allowed complex data structures like arrays and stacks to be accessed more easily. This improved performance and gave programmers more flexibility in writing assembly programs.

2.

The 8086 is considered the foundation of modern x86 processors because it introduced the **x86 instruction set architecture (ISA)**. Modern Intel and even AMD processors still maintain backward compatibility with this original instruction set.

In my opinion, what makes it foundational is that future processors like the 80286, 80386, Pentium, and even modern Core series extended the same architecture instead of replacing it. This means software written decades ago can still run on modern CPUs because of backward compatibility.

I think this design decision shaped the entire PC industry. When IBM selected an 8086-based processor for early PCs, it standardized the x86 platform. Because of that, x86 became dominant in desktops and laptops, influencing operating systems like Windows and Linux.

3.

Strengths

1. Expanded Memory Access (1MB Address Space)

The segmented memory model allowed access to 1MB of RAM, which was a big improvement compared to previous 8-bit processors limited to 64KB. I think this made the 8086 suitable for more advanced applications and operating systems.

2. Rich Instruction Set

The 8086 had a relatively powerful and flexible instruction set. It supported string operations, arithmetic, logical operations, and stack management. This made it more versatile for software development.

Limitations

1. Segmented Memory Complexity

Although segmentation expanded memory, it also made programming more complicated. I think managing segment registers increased the chances of programming errors and made memory handling less straightforward.

2. Lack of Built-in Floating-Point Unit (FPU)

The 8086 did not have an integrated floating-point unit. Complex

mathematical operations required a separate coprocessor (like the 8087).
This limited performance in scientific and engineering applications.

References:

GeeksforGeeks. (2025, September 12). *Addressing modes in 8086 microprocessor*.
<https://www.geeksforgeeks.org/computer-organization-architecture/memory-segmentation-8086-microprocessor/>

GeeksforGeeks. (2025, September 12). *Architecture of 8086*.
<https://www.geeksforgeeks.org/computer-organization-architecture/architecture-of-8086/>

GeeksforGeeks. (2025, September 12). *Addressing modes in 8086*.
<https://www.geeksforgeeks.org/addressing-modes/>

Tpoint Tech. (n.d.). *8086 microprocessor: Addressing modes*.
<https://www.tpointtech.com/8086-microprocessor>

General History & x86 Foundation

Wikipedia contributors. (2026, February). *Intel 8086*. Wikipedia.
https://en.wikipedia.org/wiki/Intel_8086

Wasil Zafar. (2026). *x86 Assembly Series Part 2: x86 CPU architecture overview*.
<https://www.wasilzafar.com/pages/series/assembly-mastery/asm-part02-cpu-architecture.html>